From HR data to actionable risk insights

Employee Risk Prediction using Support Vector Machines (SVM).

Machine Learning | HR Analytics | Operations & Continuous Improvement

The problem: Employee risk is rarely driven by a single factor

- Employee disengagement, low performance, or potential attrition rarely appear suddenly or in isolation.
- Risk usually emerges from the interaction between:
  1. individual behavior,
  2. operational conditions,
  3. and organizational structure.
- Simple rule-based approaches fail to capture this complexity.

Employee risk is a systemic phenomenon, not an individual flaw.

Why this is NOT a linear problem

The same metric does not imply the same risk

- Two employees with the same punctuality can show completely different risk outcomes.
- High productivity does not always offset:
  1. low engagement,
  2. unstable contracts,
  3. or high operational stress.
- Risk does not appear at fixed thresholds,
  but inside specific combinations of conditions.

Risk appears in regions of the feature space, not in simple if–then rules.

Dataset Overview | HR, 786 employee snapshots

Feature groups:

- **Personal factors**
  Training, punctuality, productivity, engagement, experience
- **Operational factor**
  Scrap / rework exposure
- **Structural factors**
  Area, shift, contract type, area rotation rate

In real organizations, these variables are typically built by integrating multiple systems:

Learning Management Systems (LMS)
→ Training hours and certifications

Time & Attendance systems
→ Punctuality and attendance ratios

Performance management systems
→ Productivity and evaluation scores

Engagement & pulse surveys
→ Engagement indicators

Quality / MES systems
→ Scrap and rework association

HR master data & analytics
→ Area, shift, contract type, rotation rates

The value of the dataset lies in how variables interact, not in their individual distributions.

Modeling Approach (Why SVM)

Support Vector Machines for non-linear HR risk patterns

- Supervised binary classification (High Risk vs No Risk)
- Support Vector Machine with RBF kernel
- Feature scaling and categorical encoding applied
- Hyperparameter tuning using cross-validation

Why SVM fits this problem

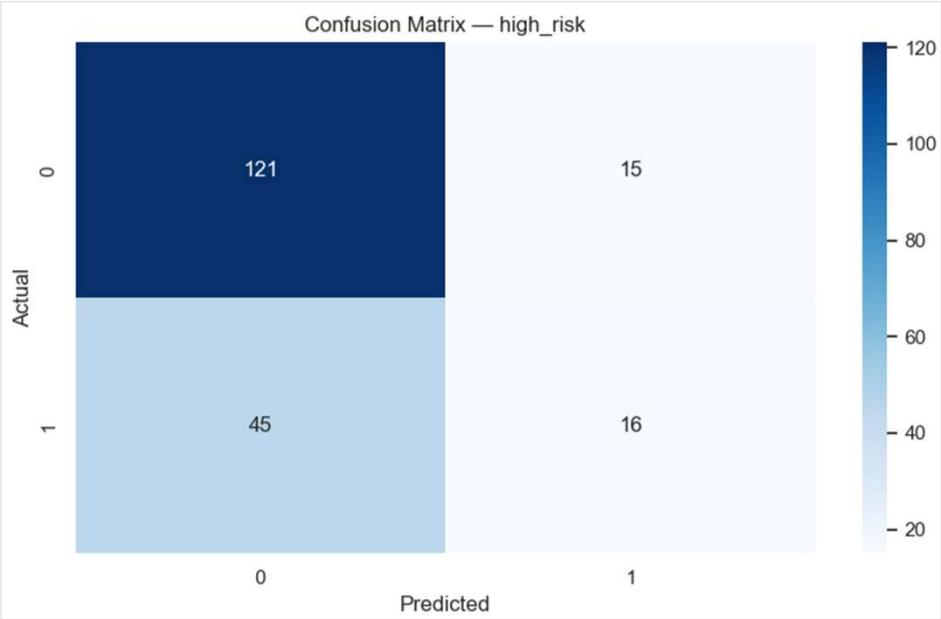
- Captures **non-linear decision boundaries**
- Learns **risk regions**, not fixed thresholds
- Effective when multiple factors interact simultaneously

The model learns where risk emerges, not just which variable matters most.

Model Performance

Balanced performance on unseen data

- Binary classification: High Risk vs No Risk
- Evaluation focused on balanced metrics:
  1. Accuracy
  2. Precision
  3. Recall
  4. F1-score
- Performance validated using a held-out test set



Confusion Matrix

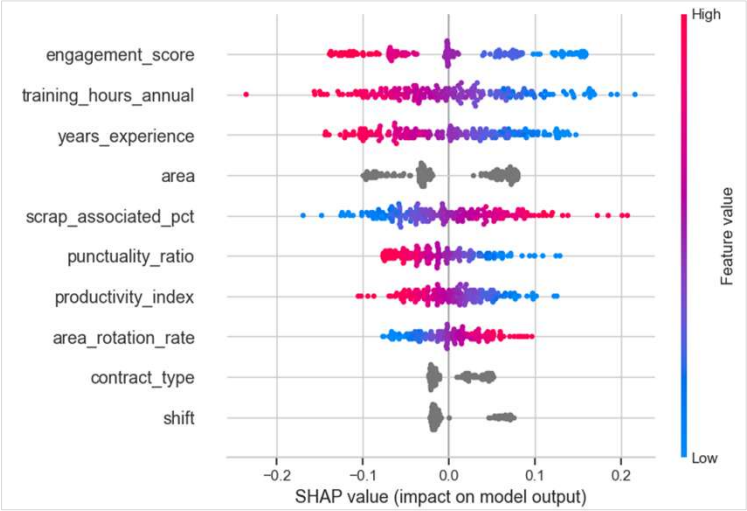
- Correct identification of most high-risk cases
- Controlled false positives to avoid over-flagging employees

The goal is not perfect prediction, but reliable risk signal detection.

Model Interpretability (SHAP)

Understanding what pushes risk up or down

- SHAP values explain predictions as additive contributions
- Each feature “pushes” the model toward:
  1. higher risk, or
  2. lower risk
- Enables transparent discussion of model behavior



Key insights

- Low engagement strongly increases risk
- Poor punctuality amplifies risk signals
- Higher scrap exposure raises operational stress
- Structural conditions can outweigh individual performance

Risk drivers are often structural, not purely personal.

Decision Regions & Scenarios

Risk as a region, not a label

- Employees fall into risk regions defined by combinations of conditions
- Two individuals with similar performance can belong to different risk regions
- Risk is contextual, not a permanent attribute

What-if scenarios

- What if training hours increase?
- What if engagement improves?
- What if contract conditions stabilize?
- What if shift assignment changes?

The model supports intervention and prevention, not judgment.

Decision Regions & Scenarios

Example A:

```
example_low_risk = {
    "training_hours_annual": 50,
    "punctuality_ratio": 0.99,
    "productivity_index": 110,
    "scrap_associated_pct": 2,
    "engagement_score": 5,
    "years_experience": 10,
    "area_rotation_rate": 0.05,
    "area": "Quality",
    "shift": "Morning",
    "contract_type": "Permanent"
```

Example B:

```
example_high_risk = {
    "training_hours_annual": 5,
    "punctuality_ratio": 0.78,
    "productivity_index": 80,
    "scrap_associated_pct": 10,
    "engagement_score": 1,
    "years_experience": 1,
    "area_rotation_rate": 0.30,
    "area": "Production",
    "shift": "Night",
    "contract_type": "Outsourcing"
```

```
Example A (expected low risk): (0, 0.011088291285074789)
Example B (expected high risk): (1, 0.9968397804692891)
```

Final reflection

What this project demonstrates

- Employee risk is a **systemic phenomenon**, not an individual flaw
- Structural and operational factors can outweigh personal performance
- Non-linear models like **SVM** are effective for complex HR risk patterns
- Interpretability is essential for responsible HR analytics

Final message

- Models should **support intervention**, not automate decisions
- Data-driven insights work best when combined with human judgment
- HR analytics is most valuable when it helps **improve systems**, not punish people

Take it further

Feel free to download the full project:
the dataset, the notebook, and the simulator.

Change the data.
Break the code.
Ask different questions.

This model is just one snapshot in an ongoing process.
And that process never truly stops.

Because improvement has no finish line.